

Events, Async, and Distributed Coordination

SSE, Redis streams, background work, leases, and multi-worker behavior.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Coordination model

Long-running workflows execute beyond request latency. A background manager and Redis leases coordinate ownership. `RunEventBus` writes bounded replay records and publishes live updates; clients reconnect with an event cursor. MongoDB remains durable history.

- Lost ownership expires through leases.
- Duplicate starts/resumes are constrained by operation and token semantics.
- SSE reconnects replay recent events then reconcile with durable runs.
- Redis loss affects coordination but does not redefine persisted history.

File	Responsibility
<code>app/runtime/coordination.py</code>	Small Redis coordination primitives shared by Uvicorn workers.
<code>app/runtime/events.py</code>	Bounded, session-scoped pub/sub for workflow Server-Sent Events.
<code>app/runtime/node_events.py</code>	Helpers for live event emission in the workflow runtime.
<code>app/workflow/orchestration.py</code>	Shared execution bookkeeping for a single workflow run.